

Project #4: Adaptive Game-Based Robotic Glove for Hand and Finger Rehabilitation

Focused on children with cerebral palsy, with adjustable use for adults after stroke

Project Aim. Design and validate a lightweight, safe and affordable robotic glove that assists finger flexion and extension, supports functional hand training, and increases motivation through interactive games. The design is optimized for children with cerebral palsy, while modular finger sleeves, adjustable straps and configurable software limits allow adaptation for adult stroke rehabilitation.

1. Project Description

The proposed system combines an adjustable wearable glove, independent finger actuators, a mirror/data glove, portable embedded control, interactive rehabilitation software, voice guidance and connected performance monitoring. It addresses common limitations in grasping, releasing, pinching, finger opposition and object manipulation. The glove will support right- or left-hand use and will be designed for quick wearing and removal by the patient, parent or therapist. Training modes will include passive, active-assistive, mirror, fine-motor, opposition, pinch, stretching and game-based training. Nine configurable assistance levels will adjust actuator effort, motion speed, range of motion and flexion/extension hold time. Independent finger channels will enable whole-hand or selective-finger exercises. Safety will be maintained through motion limits, current/force monitoring, watchdog logic, quick release and emergency stop functions.

2. Project Objectives

- Develop a modular, adjustable pediatric-first glove that is easy to wear and remove.
- Provide right/left configurations and interchangeable finger-size components.
- Implement flexion, extension, pinch, opposition and selective-finger assistance.
- Provide nine assistance levels with configurable speed, range and hold time.
- Develop mirror training using a sensor glove on the less-affected hand.
- Create age-appropriate games, voice prompts and motivational feedback.
- Measure movement quality, repetitions, task score and progress.
- Enable secure PC/mobile follow-up for clinicians and parents.

Robotic Glove-Assisted Hand Rehabilitation for Children with Cerebral Palsy



Figure 1. Proposed project concept, training modes, software assessment and follow-up workflow.

3. Proposed System Layout

The embedded controller maintains the local closed-loop safety system. The PC, tablet or mobile application provides games, voice guidance, assessment and reporting, while the IoT layer synchronizes approved session data for clinician and parent follow-up.

Proposed System Layout, Sensing, Control, Software and IoT Information Flow

A child-adaptable robotic glove platform for hand rehabilitation in children with cerebral palsy and adults after stroke.

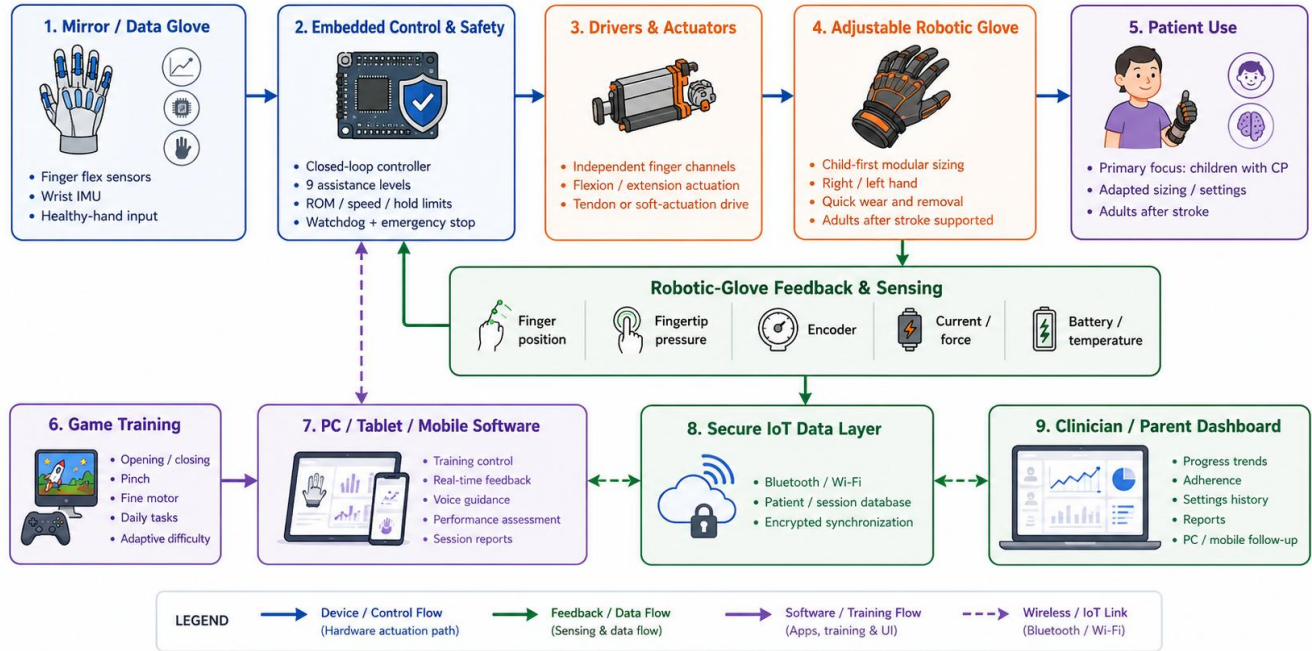


Figure 2. System layout, sensing, control, software and IoT information flow.

4. Sensors, Actuation and Safety

Subsystem	Sensors / Hardware	Function
Mirror/data glove	Finger-flex sensors; wrist IMU; optional pinch/contact sensor	Captures healthy-hand motion for bilateral or individual-finger mirror training.
Robotic glove	Finger bend/position sensors; fingertip FSRs; actuator encoders	Measures motion, contact, grasp events and tracking accuracy.
Actuation feedback	Tendon-force sensor or motor-current sensing; driver temperature	Limits excessive assistance and detects resistance or blockage.
Safety and power	Limit switches; emergency stop; watchdog; battery voltage/temperature	Enforces ROM limits, rapid release and controller/battery supervision.

5. Game-Based Software and Assessment

Opening, closing, pinch, opposition and individual-finger movements will control games such as balloon popping, catching objects, shape matching, pinch targets and virtual daily-living tasks. Difficulty will adapt to the user's range and previous performance.

- Assessment: ROM, repetitions, accuracy, completion time and smoothness.
- Session score, active contribution, assistance level and adherence.
- Voice broadcast for instructions, encouragement and mode changes.
- Automatic progress graphs and session summaries.

6. IoT Follow-Up and Deliverables

Bluetooth or Wi-Fi will connect the controller to PC, tablet and mobile software. Approved records will be stored locally and synchronized to a secure database. The dashboard will show progress, completed sessions, assistance history and downloadable reports. All actuation commands remain under local safety control.

- Adjustable robotic glove and mirror/data glove prototypes.
- Embedded controller, actuator drivers and safety circuit.
- Interactive games, voice guidance and assessment software.
- PC/mobile dashboard, IoT database and progress reports.
- CAD, electronics, code, testing protocol and demonstration video.

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